

Cyber & Information Security

Practical No 8

Aim: Analyze and identify malware samples using antivirus tools, analyze their behavior, and develop countermeasures to mitigate their impact

Step 1: Update Ubuntu

Open the Terminal and run:

```
sudo apt update
```

```
sudo apt upgrade -y
```

```
user@ubuntu:~$ sudo apt update
[sudo] password for user:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Some packages can be upgraded, but 'apt list --upgradable' to see them.
user@ubuntu:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau libdrm-radeon libgl1-amdgpu-dri libglapi-mesa libllvm7t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following packages have been kept back:
  libgl1-amdgpu-dri libglapi-mesa
The following packages will be upgraded:
  curl libcurl3t64-gnutls libcurl3t64
3 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
3 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Need to get 88 kB of archives.
After this operation, 3072 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu0.13 [228 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64 amd64 8.5.0-2ubuntu0.13 [343 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu0.13 [88 kB]
Fetched 884 kB in 2s (400 kB/s)
(Reading database ... 47687 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu0.13_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu0.13) over (8.5.0-2ubuntu0.12) ...
Preparing to unpack .../libcurl3t64_8.5.0-2ubuntu0.13_amd64.deb ...
Unpacking libcurl3t64 (8.5.0-2ubuntu0.13) over (8.5.0-2ubuntu0.12) ...
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu0.13_amd64.deb ...
Unpacking libcurl3t64-gnutls (8.5.0-2ubuntu0.13) over (8.5.0-2ubuntu0.12) ...
Setting up libcurl3t64-gnutls (8.5.0-2ubuntu0.13) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu0.13) ...
Setting up curl (8.5.0-2ubuntu0.13) ...
Processing triggers for man-db (2.12.0-4ubuntu0) ...
Processing triggers for libc-bin (2.39-0ubuntu0) ...
user@ubuntu:~$
```

Step 2: Install Snort

Install Snort using:

```
sudo apt install snort -y
```

```
Select user@D0C7XP-PBEE68-0:~$ sudo apt install smart-y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
libidn2-0 libidn2-radius libidn2-mass libidn2-res libidn2-drif-d
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
libauthen-perl perl libbz2-perl libbz2-dev perl libbz2-fork-perl libbz2-panner-perl libbz2-tagset-perl
libbz2-tree-perl libbz2-cookies-perl libbz2-daemon-perl libbz2-date-perl libbz2-message-perl
libbz2-negotiate-perl libbz2-hmail-perl libbz2-socket-sal-perl libbz2jit-5.1-2 libbz2jit-5.1-common
libbz2-mediaexec-perl libbz2-protocol-Atpz-perl libbz2-tls-perl libbz2-vcp-perl libbz2-wpa-sal-perl
libbz2-xdmp-perl libbz2-filter-qames libbz2filter libbz2 libbz2date-perl libbz2-tiny-perl libbz2-libari-perl
libbz2-perl libbz2-robofilters-perl net-tools oidsmaster perl-spread-defaults smart-common smart-common-libraries
smart-rules-default
Suggested packages:
libbz2-doc perl libbz2-dev libbz2-fork-perl libbz2-panner-perl libbz2-tagset-perl
libbz2-tree-perl libbz2-cookies-perl libbz2-daemon-perl libbz2-date-perl libbz2-message-perl
libbz2-negotiate-perl libbz2-hmail-perl libbz2-socket-sal-perl libbz2jit-5.1-2 libbz2jit-5.1-common
libbz2-mediaexec-perl libbz2-protocol-Atpz-perl libbz2-tls-perl libbz2-vcp-perl libbz2-wpa-sal-perl
libbz2-xdmp-perl libbz2-filter-qames libbz2filter libbz2 libbz2date-perl libbz2-tiny-perl libbz2-libari-perl
libbz2-perl libbz2-robofilters-perl net-tools oidsmaster perl-spread-defaults smart-common smart-rules-default
net-tools 4.9 only installed, 0 to remove and 2 not upgraded.
Need to get 445 kB of archives.
After this operation, 17.3 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbz2jit-5.1-2-common all 2.1.0-glibc20231223.c525bcb+dfsg-1ubuntu0.1 [40.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbz2jit-5.1-2-amd64 2.1.0-glibc20231223.c525bcb+dfsg-1ubuntu0.1 [276 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbz2-dev amd64 2.1.0-15build1 [248 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 smart-common-libraries amd64 2.9.20-8deb1ubuntu1 [899 B]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 smart-rules-default all 2.9.20-8deb1ubuntu1 [58 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 smart-common all 2.9.20-8deb1ubuntu1 [47.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe/main amd64 net-tools amd64 2.10-0ubuntu4.4 [284 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbz2-dev amd64 1.17.0-1ubuntu2 [30.7 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 libbz2filter amd64 1.0-2-build1 [16.4 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbz2filter-qaes amd64 1.0-2-ubuild1 [47.7 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/universe amd64 libbz2tagset amd64 2.0.7-5-1build1 [92.9 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 smart amd64 2.9.20-8deb1ubuntu1 [791 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/main amd64 libbz2-perl amd64 0.40-build1 [48.7 kB]
```

Step 3: Check the Network Interface

Use:

ip address

```
user@DESKTOP-P8BQDQ6:~$ ip address
Processing triggers for man-db (2.32.0-4ubuntu1) ...
user@DESKTOP-P8BQDQ6:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: bond0: <BROADCAST,MULTICAST,MASTER> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 4a:27:34:fc:c8:0a brd ff:ff:ff:ff:ff:ff
3: dummy0: <BROADCAST,NOARP> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 92:7b:57:bd:82:d5 brd ff:ff:ff:ff:ff:ff
4: tunl0@vnet: <NOARP> mtu 1400 qdisc noop state DOWN group default qlen 1000
    link/ipip 0.0.0.0 brd 0.0.0.0
5: sit0@vnet: <NOARP> mtu 1400 qdisc noop state DOWN group default qlen 1000
    link/sit 0.0.0.0 brd 0.0.0.0
6: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:39:16:00 brd ff:ff:ff:ff:ff:ff
    inet 172.29.188.118/20 brd 172.29.191.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe29:1800/64 scope link
        valid_lft forever preferred_lft forever
user@DESKTOP-P8BQDQ6:~$
```

Step 4: Test Snort Configuration

Run:

sudo snort -T -c /etc/snort/snort.conf

```
user@DESKTOP-P8BQDQ6:~$ sudo snort -T -c /etc/snort/snort.conf
Verifying Preprocessor Configurations
WARNING: Flowbits key 'snb_tree_create_llsrc' is set but not ever checked.
WARNING: Flowbits key 'no_sql_seen_dns' is checked but not ever set.
33 out of 1024 flowbits in use.

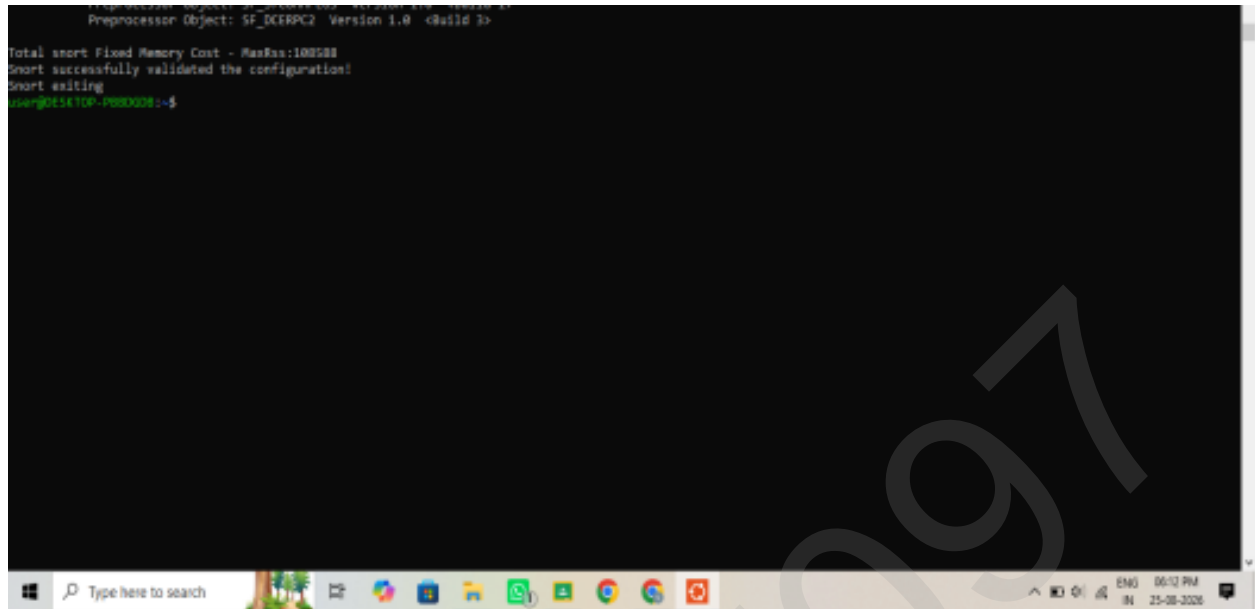
Snort at the end of rules:65004

[ Port Based Pattern Matching Memory ]
-----
[ Aho-Corasick Summary ] -----
Storage Format      : Full-Q
Finite Automaton    : DFA
Alphabet Size       : 256 chars
Sliced State       : Variable (1,2,4 bytes)
Instances          : 215
  1 byte states    : 204
  2 byte states    : 11
  4 byte states    : 0
Characters         : 64765
States             : 31951
Transitions        : 863868
State Density       : 10.65
Patterns           : 5842
Match States       : 3836
Memory (MB)        : 16.00
  Patterns         : 0.51
  Match Lists      : 2.02
DFA
  1 byte states    : 1.02
  2 byte states    : 11.96
  4 byte states    : 0.00
-----
[ Number of patterns truncated to 20 bytes: 1000 ]
Snort at the end of detection rules:166508

--- Initialization Complete ---

-#- Snort! -#-
o-#-
o-#-
Version 2.9.20 GRC (Build 82)
By Martin Roesch & The Snort team: http://www.snort.org/contact/team
Copyright (C) 2004-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 2000-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TRACET_V5)
Using OpenSSL version: 3.30 2023-06-16
```

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A screenshot of a Windows terminal window. The terminal displays the output of a Snort configuration check. The text shown is: "Preprocessor Object: SF_DKRPC2 Version 1.0 (Build 1)", "Total snort Fixed Memory Cost - MaxRes:100588", "Snort successfully validated the configuration!", "Snort exiting", and a prompt "root@kali:~#". The terminal window has a taskbar at the bottom with various application icons and a system tray showing the date and time as 25-08-2020, 06:12 PM.

Step 5: Start Snort in Network Monitoring Mode

Run:

```
sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
```

Replace enp0s3 with your actual network interface.

- -i specifies the network interface.
- -A console displays alerts on the terminal.
- -c specifies the Snort configuration file.

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```
user@DESKTOP-P8BQGB:~$ sudo snort -l /etc/snort/snort.conf -A console -c /etc/snort/snort.conf
Running in IDS mode

---- Initializing Snort ----
Initializing Output Plugin!
Initializing Preprocessor!
Initializing Plug-In!
Parsing Rules File "/etc/snort/snort.conf"
ERROR: /etc/snort/snort.conf(0) Unable to open rules file "/etc/snort/snort.conf": No such file or directory.

Fatal Error, Quitting..
user@DESKTOP-P8BQGB:~$ sudo snort -l /etc/snort/snort.conf -A console -c /etc/snort/snort.conf
Running in IDS mode

---- Initializing Snort ----
Initializing Output Plugin!
Initializing Preprocessor!
Initializing Plug-In!
Parsing Rules File "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 811 883 991 993 994 1230 1414 1741 1830 2301 2381 2889 3037 3128 3702 4343 4843 5250 6888 7060:7061 7144:7145 7510 7777 7779 8080 8080 8081 8082 8083 8084 8085 8086 8087 8088 8089 8123 8180 8181 8243 8260 8300 8585 8899 9000 9060 9060:9091 9443 9999 11371 14443:14444 41000 50002 55555 ]
PortVar 'SMTP_PORTS' defined : [ 25 ]
PortVar 'IMAP_PORTS' defined : [ 143 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5060 ]
PortVar 'FILE_DATA_PORTS' defined : [ 99:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2889 3037 3128 3702 4343 4848 5250 6500 7060:7061 7144:7145 7510 7777 7779 8080 8082 8084 8085 8086 8087 8088 8089 8123 8123 8180:8181 8243 8260 8300 8585 8899 9000 9060 9060:9091 9443 9999 11371 14443:14444 41000 50002 55555 55 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-0
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
  Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsdf_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
Finished loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsdf_dce2_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsdf_s7complog_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsdf_reputation_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsdf_fuzzylook_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsdf_gtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsdf_ssl_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsdf_modbus_preproc.so... done
```

Step 6: Create a Custom IDS Rule

Open the local rules file:

```
sudo nano /etc/snort/rules/local.rules
```

Add:

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1);
```

This rule generates an alert whenever ICMP ping traffic is detected.

```
user@DESKTOP-P8BQGB:~$ nano /etc/snort/rules/local.rules
/etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/25 10:15:43 mac_fap $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1);
```

Step 7: Ensure the Local Rule File Is Included

Open:

`sudo nano /etc/snort/snort.conf`

Check that this line is present:

`include $RULE_PATH/local.rules`



```
user@DESKTOP-P86Q206: ~
$ nano /etc/snort/snort.conf

#####
# Step 47: Customize your rule set
# For more information, see Snort Manual, Writing Snort Rules
#
# NOTE: All categories are enabled in this conf file
#####

# Note to Debian users: The rules preinstalled in the system
# can be "very" out of date. For more information please read
# the /usr/share/doc/snort-rules-default/README.Debian file
#
# If you install the official VRT Sourcefire rules please review this
# configuration file and re-enable (remove the comment in the first line) those
# rules files that are available in your system (in the /etc/snort/rules
# directory)

# site specific rules
include $RULE_PATH/local.rules

# The include files commented below have been disabled
# because they are not available in the stock Debian
# rules. If you install the Sourcefire VRT please make
# sure you re-enable them again:

#include $RULE_PATH/app-detect.rules
#include $RULE_PATH/attack-responses.rules
#include $RULE_PATH/backdoor.rules
#include $RULE_PATH/bad-traffic.rules
#include $RULE_PATH/bajacklist.rules
#include $RULE_PATH/botnet-cnc.rules
#include $RULE_PATH/browser-chrome.rules
#include $RULE_PATH/browser-firefox.rules
#include $RULE_PATH/browser-ie.rules
#include $RULE_PATH/browser-other.rules
#include $RULE_PATH/browser-plugins.rules
#include $RULE_PATH/browser-webkit.rules
#include $RULE_PATH/chat.rules
```

Step 8: Test the Configuration Again

Run:

`sudo snort -T -c /etc/snort/snort.conf`

Correct any errors before continuing.

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```
snort@DESKTOP-P880208:~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Patterns : 0.51
Match Lists : 1.01
DFA
1 byte states : 1.02
2 byte states : 13.96
4 byte states : 0.00
-----
Number of patterns truncated to 20 bytes: 1038 ]
MaxRets at the end of detection rules:100000

---- Initialization Complete ----

/* Snort! */
Version 2.9.20 GRC (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact@team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_SDP Version 1.1 <Build 1>
Preprocessor Object: SF_DNP4 Version 1.1 <Build 1>
Preprocessor Object: SF_DDP Version 1.0 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: apqids Version 1.1 <Build 5>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_HOODS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPFLINT Version 1.2 <Build 12>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SPCOMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCEKPC2 Version 1.0 <Build 3>

Total snort fixed memory cost - MaxRets:100000
Snort successfully validated the configuration!
Snort exiting
snort@DESKTOP-P880208:~$ ip addr
```

Step 9: Start IDS Monitoring

Run:

```
sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
```

The IDS is now actively monitoring network traffic.

```
snort@DESKTOP-P880208:~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Patterns : 0.51
Match Lists : 1.01
DFA
1 byte states : 1.02
2 byte states : 13.96
4 byte states : 0.00
-----
Number of patterns truncated to 20 bytes: 1038 ]
snort (DAQ configured to passive).
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x7f8c0a02dc0 (128K)
Decoding Ethernet

---- Initialization Complete ----

/* Snort! */
Version 2.9.20 GRC (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact@team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_SDP Version 1.1 <Build 1>
Preprocessor Object: SF_DNP4 Version 1.1 <Build 1>
Preprocessor Object: SF_DDP Version 1.0 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: apqids Version 1.1 <Build 5>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_HOODS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPFLINT Version 1.2 <Build 12>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SPCOMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DCEKPC2 Version 1.0 <Build 3>

Commencing packet processing (pid=1287)
```

10. Step 7: Ensure the Local Rule File Is Included

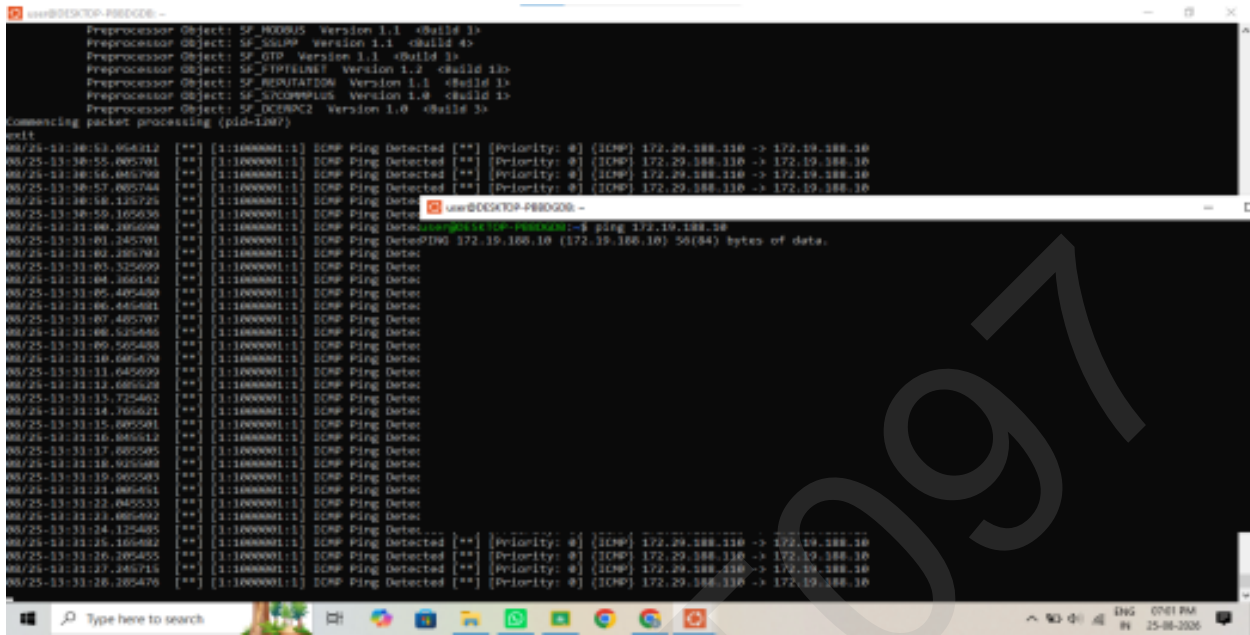
Open:

```
sudo nano /etc/snort/snort.conf
```

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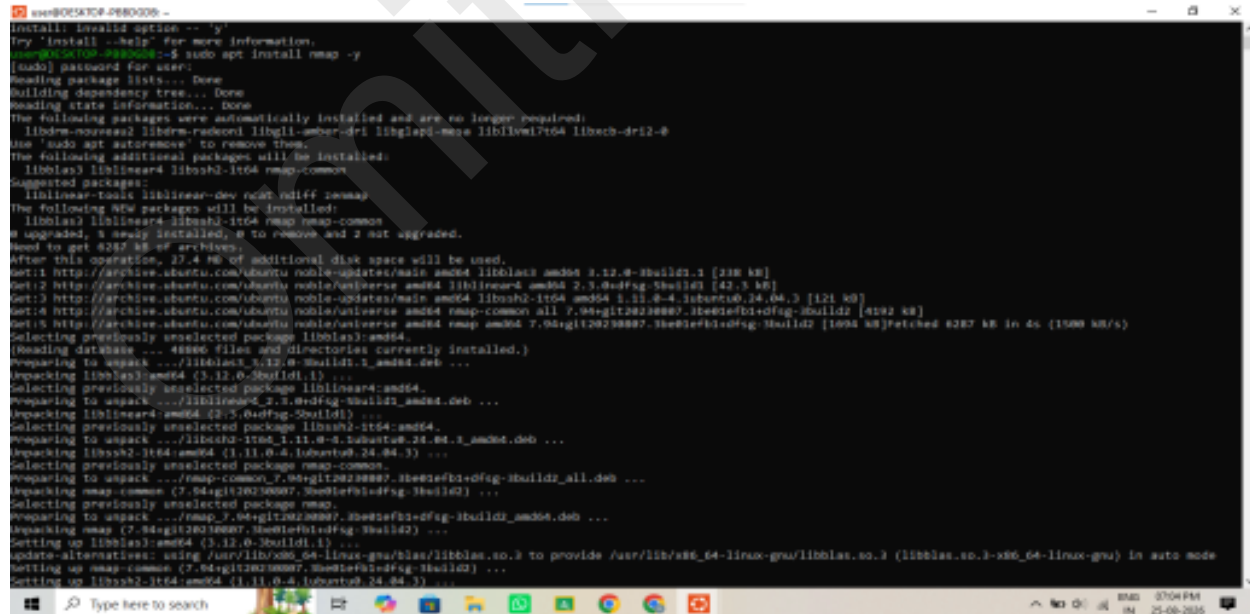
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Step 11: Test Port Scanning Detection

Install Nmap:

```
sudo apt install nmap -y
```



Add the following rule to `/etc/snort/rules/local.rules`:

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alert tcp any any -> \$HOME_NET any (flags:S; msg:"Possible TCP SYN Scan Detected";

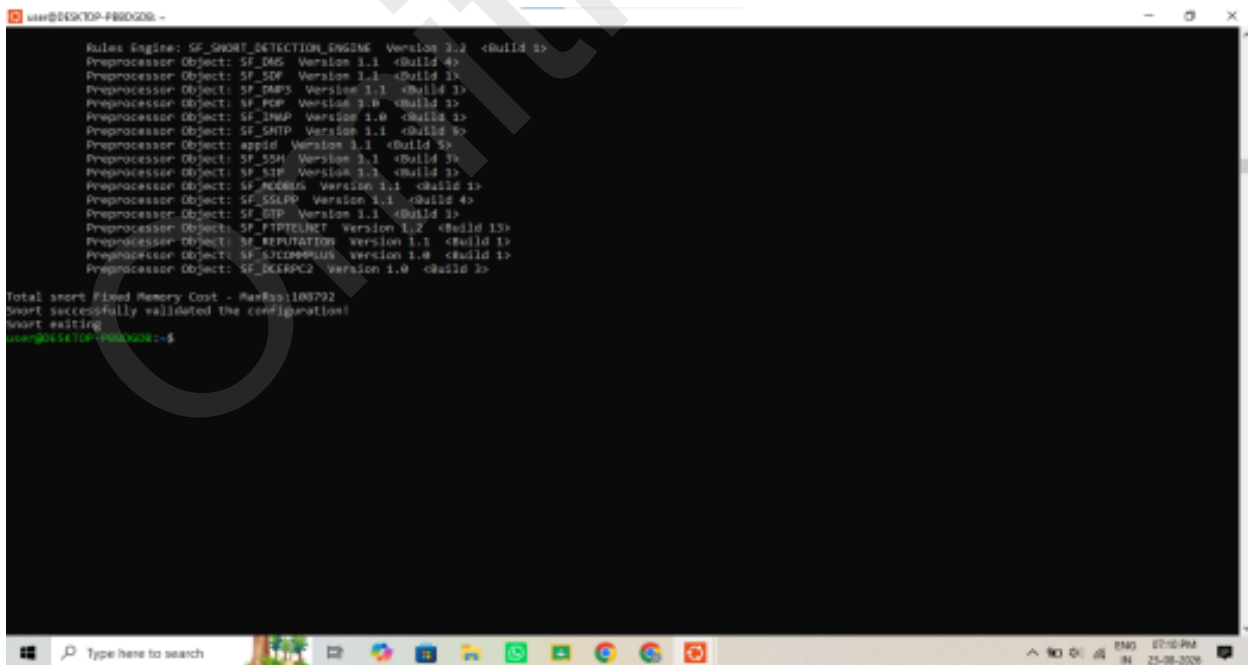
sid:1000002; rev:1;)

Restart Snort. From an authorized lab machine:

nmap -sS <TARGET-IP-ADDRESS>

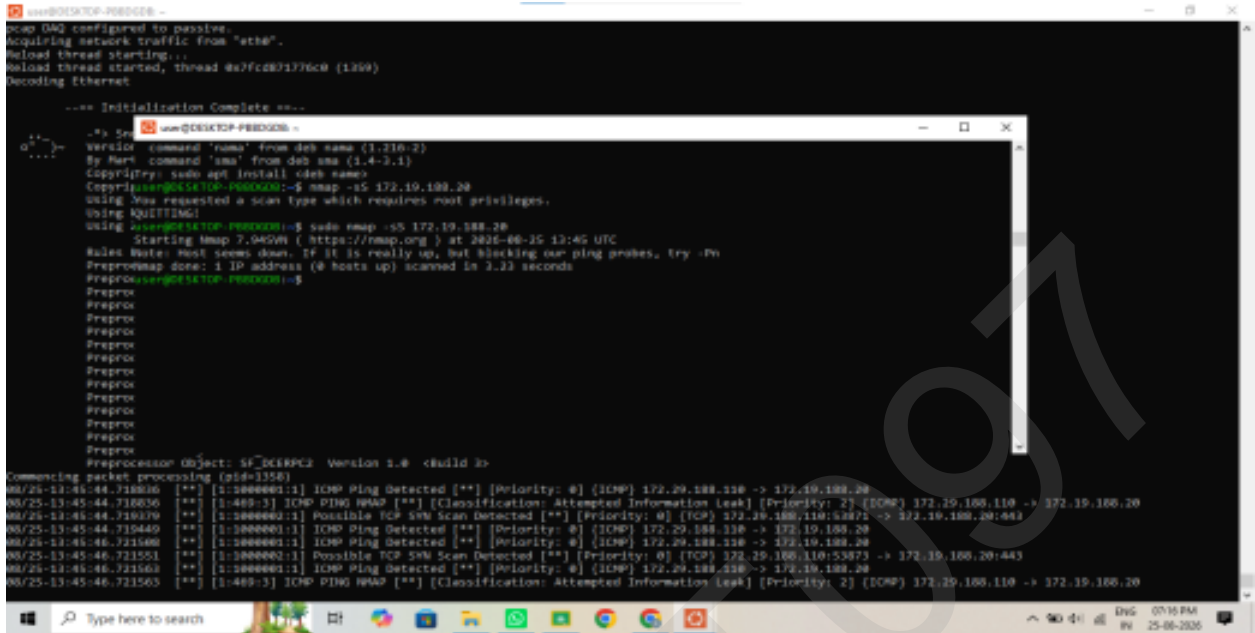


```
user@DESKTOP-P88DQ26: ~  
C:\Program Files\Snort\bin> nano /etc/snort/rules/local.rules  
# $Id: local.rules,v 1.11 2004/07/23 20:15:08 dmc isp $  
#-----  
# LOCAL RULES  
#-----  
# This file intentionally does not come with signatures. Put your local  
# additions here.  
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)  
alert tcp any any -> $HOME_NET any (flags:S; msg:"Possible TCP SYN Scan Detected"; sid:1000002; rev:1;)  
File name to write: /etc/snort/rules/local.rules  
[Help] [Format] [Append] [Backup File]  
[Cancel] [Quit] [Format] [Browse]
```



```
user@DESKTOP-P88DQ26: ~  
C:\Program Files\Snort\bin> snort -t  
Rules Engine: SF_SHORT_DETECTION_ENGINE Version 3.0 <Build 1>  
Preprocessor Object: SF_DNS Version 1.1 <Build 4>  
Preprocessor Object: SF_SDF Version 1.1 <Build 1>  
Preprocessor Object: SF_DAP Version 1.1 <Build 1>  
Preprocessor Object: SF_PDP Version 1.0 <Build 1>  
Preprocessor Object: SF_INMP Version 1.0 <Build 1>  
Preprocessor Object: SF_SHTP Version 1.1 <Build 1>  
Preprocessor Object: apd Version 1.1 <Build 1>  
Preprocessor Object: SF_SSH Version 1.1 <Build 1>  
Preprocessor Object: SF_SIP Version 1.1 <Build 1>  
Preprocessor Object: SF_ICM2 Version 1.1 <Build 1>  
Preprocessor Object: SF_SSLDP Version 1.1 <Build 4>  
Preprocessor Object: SF_GTP Version 1.1 <Build 1>  
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>  
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>  
Preprocessor Object: SF_CIRCPPUS Version 1.0 <Build 1>  
Preprocessor Object: SF_DETECTPC2 Version 1.0 <Build 1>  
Total snort /lib Memory Cost - 164831106792  
Snort successfully validated the configuration!  
snort exiting  
user@DESKTOP-P88DQ26: ~
```

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Step 12: View IDS Alerts

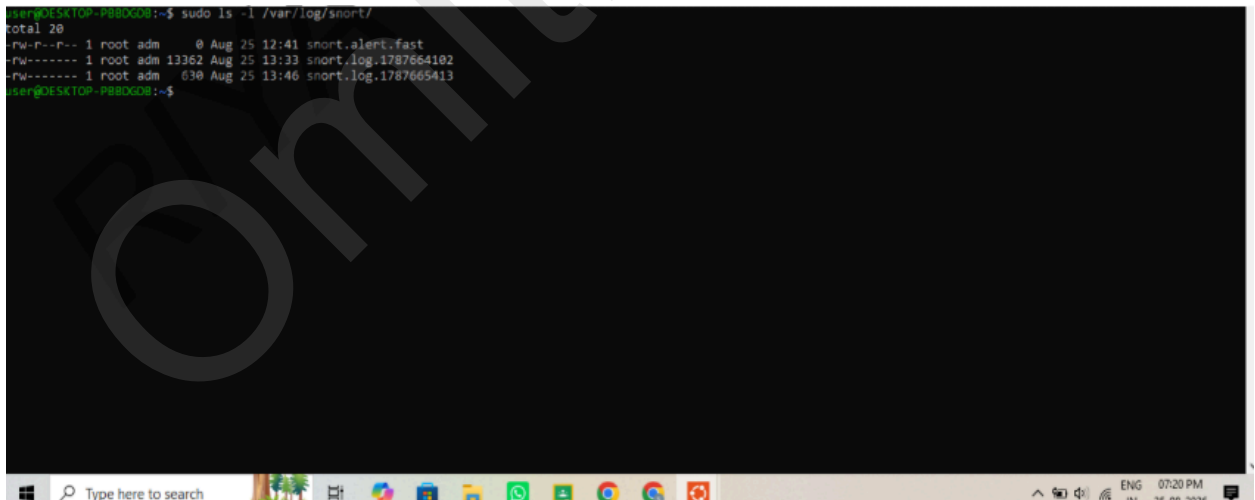
Alerts may be displayed directly when using -A console. Logs can also be checked in:

```
/var/log/snort/
```

Commands:

```
sudo ls /var/log/snort/
```

```
sudo cat /var/log/snort/alert
```



Step 13: Monitor Traffic Using Wireshark

Omith Thilakan

T097

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sudo apt install wireshark -y

```
user@DESKTOP-P8BDDGB: ~$ sudo apt install wireshark -y
Get:73 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-icccm4 amd64 0.4.1-1.1build3 [10.8 kB]
Get:74 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-util1 amd64 0.4.0-1build3 [10.7 kB]
Get:75 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-image0 amd64 0.4.0-2build1 [10.8 kB]
Get:76 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-keysyms1 amd64 0.4.0-1build4 [7956 B]
Get:77 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-render-util0 amd64 0.3.9-1build4 [9608 B]
Get:78 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-shape0 amd64 1.15-1ubuntu2 [6108 B]
Get:79 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-xkb1 amd64 1.15-1ubuntu2 [32.3 kB]
Get:80 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcb-composite0 amd64 1.6.0-1build1 [14.5 kB]
Get:81 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6gui6 amd64 6.4.2+dfsg-21.1build5 [3119 kB]
Get:82 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6network6 amd64 6.4.2+dfsg-21.1build5 [740 kB]
Get:83 http://archive.ubuntu.com/ubuntu noble/universe amd64 libswscale7 amd64 7:6.1.1-3ubuntu5 [193 kB]
Get:84 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6multimedia6 amd64 6.4.2-1build3 [527 kB]
Get:85 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6opengl6 amd64 6.4.2+dfsg-21.1build5 [420 kB]
Get:86 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6widgets6 amd64 6.4.2+dfsg-21.1build5 [2648 kB]
Get:87 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6printsupport6 amd64 6.4.2+dfsg-21.1build5 [222 kB]
Get:88 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6qml6 amd64 6.4.2+dfsg-4build3 [1619 kB]
Get:89 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6qmlmodels6 amd64 6.4.2+dfsg-4build3 [267 kB]
Get:90 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6quick6 amd64 6.4.2+dfsg-4build3 [2801 kB]
Get:91 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6svg6 amd64 6.4.2-4ubuntu3 [156 kB]
Get:92 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandclient6 amd64 6.4.2-5build3 [251 kB]
Get:93 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandcompositor6 amd64 6.4.2-5build3 [427 kB]
Get:94 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandeglclientintegration6 amd64 6.4.2-5build3 [19.5 kB]
Get:95 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6waylandeglcompositorintegration6 amd64 6.4.2-5build3 [15.2 kB]
Get:96 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqt6shellintegration6 amd64 6.4.2-5build3 [38.0 kB]
Get:97 http://archive.ubuntu.com/ubuntu noble/universe amd64 libam12t64 amd64 0.4.8+dfsg2-10.1build1 [112 kB]
Get:98 http://archive.ubuntu.com/ubuntu noble/universe amd64 libspandsp2t64 amd64 0.0.6+dfsg-2.1build1 [311 kB]
Get:99 http://archive.ubuntu.com/ubuntu noble/main amd64 libspeexdsp1 amd64 1.2.1-1ubuntu3 [43.1 kB]
Get:100 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwireshark-data all 4.2.2-1.1build3 [454 kB]
Get:101 http://archive.ubuntu.com/ubuntu noble/main amd64 libsbcl amd64 2.0-1build1 [33.9 kB]
Get:102 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwireshark115t64 amd64 4.2.2-1.1build3 [121 kB]
Get:103 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwireshark14t64 amd64 4.2.2-1.1build3 [259 kB]
Get:104 http://archive.ubuntu.com/ubuntu noble/universe amd64 libwireshark17t64 amd64 4.2.2-1.1build3 [23.7 MB]
Get:105 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 mesa-va-drivers amd64 25.2.8-0ubuntu0.24.04.2 [6784 B]
Get:106 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 mesa-vdpau-drivers amd64 25.2.8-0ubuntu0.24.04.2 [23.2 kB]
Get:107 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-rtx-platformtheme amd64 6.4.2+dfsg-21.1build5 [55.1 kB]
Get:108 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-gpu-plugins amd64 6.4.2+dfsg-21.1build5 [89.3 kB]
Get:109 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-translations-l10n all 6.4.2-1 [1761 kB]
Get:110 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-wayland amd64 6.4.2-5build3 [127 kB]
Get:111 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 1965-va-driver amd64 2.4.1+dfsg1-1ubuntu0.2 [332 kB]
Get:112 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 va-driver-all amd64 2.20.0-2ubuntu0.2 [5014 B]
Get:113 http://archive.ubuntu.com/ubuntu noble/main amd64 vdpau-driver-all amd64 1.5-2build1 [4414 B]
Get:114 http://archive.ubuntu.com/ubuntu noble/universe amd64 wireshark-common amd64 4.2.2-1.1build3 [587 kB]
85% [114 wireshark-common 20.0 kB/587 kB 3%]
2235 kB/s 29
```

```
pi@raspberrypi: ~$ sudo apt install wireshark -y
Package configuration

Configuring wireshark-common

Dumpcap can be installed in a way that allows members of the 'wireshark' system group to capture packets. This is recommended over the alternative of running Wireshark/Tshark directly as root, because less of the code will run with elevated privileges.

For more detailed information, please see /usr/share/doc/wireshark-common/README.Debian.gz once the package is installed.

Enabling this feature may be a security risk, so it is disabled by default. If in doubt, it is suggested to leave it disabled.

Should non-superusers be able to capture packets?
<Yes> <No>
```